

COMP0089 Mock Examination — Solutions

Academic Year 2025–26

Question 1A — Bandits

Preliminary: action values after the data. From Table 1 (timesteps 1–5):

Action	Times pulled	Total reward	$Q_6(a)$
a_1	2	$+1 + (-1) = 0$	0
a_2	2	$+1 + (-1) = 0$	0
a_3	1	$+1$	$+1$

(a) ε -greedy with $\varepsilon = 0.3$ at $t = 6$

[3 marks]

Solution. The greedy action is a_3 since $Q_6(a_3) = 1 > 0 = Q_6(a_1) = Q_6(a_2)$.

With $\varepsilon = 0.3$ and $|\mathcal{A}| = 3$:

$$\pi_6(a_1) = \frac{0.3}{3} = 0.1, \quad \pi_6(a_2) = 0.1, \quad \pi_6(a_3) = 1 - 0.3 + \frac{0.3}{3} = 0.8.$$

(b) UCB with $c = 1$ at $t = 6$

[4 marks]

Solution. The UCB value of each action at time $t = 6$ is

$$\text{UCB}_6(a) = Q_6(a) + \sqrt{\frac{\ln 6}{N_6(a)}}.$$

Numerically, $\ln 6 \approx 1.792$:

$$\text{UCB}_6(a_1) = 0 + \sqrt{\frac{1.792}{2}} = \sqrt{0.896} \approx 0.947,$$

$$\text{UCB}_6(a_2) = 0 + \sqrt{\frac{1.792}{2}} \approx 0.947,$$

$$\text{UCB}_6(a_3) = 1 + \sqrt{\frac{1.792}{1}} \approx 1 + 1.339 = 2.339.$$

a_3 has the highest UCB value (both due to its higher Q and its larger uncertainty from having been pulled only once). UCB is deterministic:

$$\pi_6(a_1) = 0, \quad \pi_6(a_2) = 0, \quad \pi_6(a_3) = 1.$$

(c) Asymptotic UCB behaviour when all arms have $q(a) = 0$

[5 marks]

Solution. When all arms have the same true expected reward, the sample means all converge to 0 as $t \rightarrow \infty$. The UCB index then reduces to

$$\text{UCB}_t(a) \approx \sqrt{\frac{\ln t}{N_t(a)}},$$

for large t . UCB selects the action with the largest index, which is the one with the *smallest* count $N_t(a)$. This forces a round-robin-like behaviour: whenever one arm falls behind in count, its exploration bonus grows and it gets selected next.

Consequently, all three arms are pulled at the same asymptotic rate:

$$\frac{N_t(a)}{t} \longrightarrow \frac{1}{3} \quad \text{for each } a \in \{a_1, a_2, a_3\}.$$

Key insight: UCB explores *all* arms infinitely often (the exploration bonus $\sqrt{\ln t / N_t(a)} \rightarrow \infty$ whenever $N_t(a)$ is bounded while $t \rightarrow \infty$). With equal true values, no arm is “weeded out,” so the pulls distribute uniformly.

Question 1B — Linear Function Approximation

(d) True value function under the uniformly random policy [4 marks]

Solution. Under $\pi(a_L|s) = \pi(a_R|s) = 0.5$ for all s , with $\gamma = 0.5$, the Bellman equations are:

$$v(s_1) = \frac{1}{2}[0 + 0.5v(s_1)] + \frac{1}{2}[2 + 0.5v(s_2)] = 1 + 0.25v(s_1) + 0.25v(s_2), \quad (1)$$

$$v(s_2) = \frac{1}{2}[1 + 0.5v(s_1)] + \frac{1}{2}[0 + 0.5v(s_3)] = 0.5 + 0.25v(s_1) + 0.25v(s_3), \quad (2)$$

$$v(s_3) = \frac{1}{2}[1 + 0.5v(s_2)] + \frac{1}{2}[0 + 0.5v(s_3)] = 0.5 + 0.25v(s_2) + 0.25v(s_3). \quad (3)$$

From (1): $0.75v(s_1) = 1 + 0.25v(s_2) \Rightarrow v(s_1) = \frac{4+v(s_2)}{3}$.

From (3): $0.75v(s_3) = 0.5 + 0.25v(s_2) \Rightarrow v(s_3) = \frac{2+v(s_2)}{3}$.

Substituting both into (2):

$$v(s_2) = 0.5 + \frac{4+v(s_2)}{12} + \frac{2+v(s_2)}{12} = 0.5 + \frac{6+2v(s_2)}{12} = 1 + \frac{v(s_2)}{6}.$$

Solving: $\frac{5}{6}v(s_2) = 1$, so $v(s_2) = \frac{6}{5}$. Back-substituting:

$$v_\pi(s_1) = \frac{26}{15} \approx 1.733, \quad v_\pi(s_2) = \frac{6}{5} = 1.2, \quad v_\pi(s_3) = \frac{16}{15} \approx 1.067.$$

(e) Expected semi-gradient TD(0) update from s_2 [5 marks]

Solution. Features: $\phi(s_1) = [1, 0]^\top$, $\phi(s_2) = [1, 1]^\top$, $\phi(s_3) = [0, 1]^\top$. Initial $\mathbf{w} = [0, 0]^\top$, $\alpha = 0.1$, agent starts in s_2 .

With $\mathbf{w} = \mathbf{0}$ we have $v_{\mathbf{w}}(s) = 0$ for all s .

The semi-gradient TD(0) update is

$$\mathbf{w} \leftarrow \mathbf{w} + \alpha \delta_t \phi(S_t), \quad \text{where } \delta_t = R_{t+1} + \gamma \mathbf{w}^\top \phi(S_{t+1}) - \mathbf{w}^\top \phi(S_t).$$

Starting in $S_0 = s_2$ under the uniform policy:

- **With prob $\frac{1}{2}$:** $A_0 = a_L$, $R_1 = 1$, $S_1 = s_1$. $\delta_0 = 1 + 0 - 0 = 1$. $\Delta \mathbf{w} = 0.1 \cdot 1 \cdot [1, 1]^\top = [0.1, 0.1]^\top$.
- **With prob $\frac{1}{2}$:** $A_0 = a_R$, $R_1 = 0$, $S_1 = s_3$. $\delta_0 = 0 + 0 - 0 = 0$. $\Delta \mathbf{w} = [0, 0]^\top$.

The expected weight vector after one step:

$$\mathbb{E}[\mathbf{w}] = [0, 0]^\top + \frac{1}{2}[0.1, 0.1]^\top + \frac{1}{2}[0, 0]^\top = [0.05, 0.05]^\top.$$

(f) **Scalar features: convergence of TD(0)**

[6 marks]

Solution. Features: $\phi(s_1) = 1$, $\phi(s_2) = 2$, $\phi(s_3) = 3$, so $v_w(s) = w \phi(s)$.

For on-policy semi-gradient TD(0) with linear features, the expected update is $\mathbb{E}[\Delta w] = \alpha(Aw + b)$, where

$$A = \sum_s \sum_a d(s) \pi(a|s) \phi(s) [\gamma \phi(s') - \phi(s)], \quad b = \sum_s \sum_a d(s) \pi(a|s) \phi(s) R(s, a).$$

Stationary distribution. Under the uniform random policy, the transition matrix is

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 0.5 & 0.5 \end{pmatrix}.$$

By symmetry (checking $d^\top P = d^\top$), $d(s_1) = d(s_2) = d(s_3) = \frac{1}{3}$.

Computing A . Each (s, a) pair contributes $\frac{1}{6}$ weight:

s	a	s'	R	$\phi(s)[\gamma\phi(s') - \phi(s)]$	$\phi(s) \cdot R$
s_1	a_L	s_1	0	$1 \cdot [0.5 \cdot 1 - 1] = -0.5$	0
s_1	a_R	s_2	2	$1 \cdot [0.5 \cdot 2 - 1] = 0$	2
s_2	a_L	s_1	1	$2 \cdot [0.5 \cdot 1 - 2] = -3$	2
s_2	a_R	s_3	0	$2 \cdot [0.5 \cdot 3 - 2] = -1$	0
s_3	a_L	s_2	1	$3 \cdot [0.5 \cdot 2 - 3] = -6$	3
s_3	a_R	s_3	0	$3 \cdot [0.5 \cdot 3 - 3] = -4.5$	0

$$A = \frac{1}{6}(-0.5 + 0 - 3 - 1 - 6 - 4.5) = \frac{-15}{6} = -2.5, \quad b = \frac{1}{6}(0 + 2 + 2 + 0 + 3 + 0) = \frac{7}{6}.$$

Convergence. Since $A = -2.5 < 0$ (negative definite in the scalar case), the linear TD(0) system is *stable* and convergent for sufficiently small constant α (specifically $|1 + \alpha A| < 1 \Leftrightarrow \alpha < 4/5$), and for any step-size schedule satisfying $\sum \alpha_t = \infty$, $\sum \alpha_t^2 < \infty$.

Fixed point. Setting $Aw^* + b = 0$:

$$w^* = -\frac{b}{A} = \frac{7/6}{2.5} = \frac{7}{15} \approx 0.467.$$

This gives $v_{w^*}(s_1) = \frac{7}{15}$, $v_{w^*}(s_2) = \frac{14}{15}$, $v_{w^*}(s_3) = \frac{7}{5}$.

Remark. These do not match the true values $(\frac{26}{15}, \frac{6}{5}, \frac{16}{15})$. Notably the ordering is *reversed*: the true values satisfy $v(s_1) > v(s_2) > v(s_3)$ but the linear features $\phi = (1, 2, 3)$ force the approximation to be monotonically increasing. The TD fixed point is the best the linear class can do under the d -weighted projection.

Question 2 — Policy Gradient

(a) Score functions for the sigmoid policy

[3 marks]

Solution. Recall $\sigma'(x) = \sigma(x)(1 - \sigma(x))$.

For a_1 : $\log \pi_\theta(a_1) = \log \sigma(\theta)$, so

$$\frac{\partial}{\partial \theta} \log \pi_\theta(a_1) = \frac{\sigma'(\theta)}{\sigma(\theta)} = \frac{\sigma(\theta)(1 - \sigma(\theta))}{\sigma(\theta)} = 1 - \sigma(\theta) = \pi_\theta(a_2).$$

For a_2 : $\log \pi_\theta(a_2) = \log(1 - \sigma(\theta))$, so

$$\frac{\partial}{\partial \theta} \log \pi_\theta(a_2) = \frac{-\sigma'(\theta)}{1 - \sigma(\theta)} = -\sigma(\theta) = -\pi_\theta(a_1).$$

$$\boxed{\frac{\partial}{\partial \theta} \log \pi_\theta(a_1) = 1 - \sigma(\theta), \quad \frac{\partial}{\partial \theta} \log \pi_\theta(a_2) = -\sigma(\theta).}$$

(b) Update from $\theta = 2$, $A = a_2$, $R = +3$, $\alpha = 0.5$

[3 marks]

Solution. $\sigma(2) = \frac{1}{1+e^{-2}} \approx 0.8808$.

The REINFORCE update is

$$\Delta\theta = \alpha R \frac{\partial}{\partial \theta} \log \pi_\theta(a_2) = 0.5 \times 3 \times (-\sigma(2)) = -1.5 \sigma(2) \approx -1.321.$$

$$\boxed{\theta' = 2 - 1.5 \sigma(2) \approx 0.679.}$$

Policy change: $\pi_\theta(a_1)$ goes from $\sigma(2) \approx 0.881$ to $\sigma(0.679) \approx 0.664$, a decrease of $|\Delta\pi| \approx 0.217$.

(c) Update from $\theta = 0$; comparison

[5 marks]

Solution. $\sigma(0) = 0.5$.

$$\Delta\theta = 0.5 \times 3 \times (-0.5) = -0.75. \quad \theta' = -0.75.$$

Policy change: $\pi_\theta(a_1)$ goes from 0.5 to $\sigma(-0.75) = \frac{1}{1+e^{0.75}} \approx 0.321$.

$$|\Delta\pi| \approx |0.321 - 0.5| = 0.179.$$

Comparison. Starting from $\theta = 2$ gives a larger policy change (0.217) than $\theta = 0$ (0.179), despite the sigmoid being *steeper* near $\theta = 0$.

Explanation. Two competing effects:

1. The *score function magnitude* $|\partial_\theta \log \pi_\theta(a_2)| = \sigma(\theta)$ is *larger* at $\theta = 2$ (0.88 vs. 0.50), producing a bigger parameter step ($|\Delta\theta| = 1.32$ vs. 0.75).
2. The *policy sensitivity* $|d\pi/d\theta| = \sigma(\theta)(1 - \sigma(\theta))$ is *larger* at $\theta = 0$ (0.25 vs. 0.10).

In this case, effect (1) dominates: the larger parameter step from $\theta = 2$ more than compensates for the lower local sensitivity, because the step is large enough to traverse the steep central region of the sigmoid.

(d) Expected update equals zero iff $r(a_1) = r(a_2)$

[5 marks]

Solution. Let $p = \sigma(\theta) = \pi_\theta(a_1)$. The expected update is

$$\begin{aligned}\mathbb{E}\left[R(A) \frac{\partial}{\partial \theta} \log \pi_\theta(A)\right] &= p r(a_1)(1-p) + (1-p) r(a_2)(-p) \\ &= p(1-p)[r(a_1) - r(a_2)].\end{aligned}$$

Since $p = \sigma(\theta) \in (0, 1)$ for all finite θ , the factor $p(1-p) > 0$. Therefore

$$\boxed{\mathbb{E}[\Delta\theta] = 0 \iff r(a_1) = r(a_2).}$$

Intuitively, the expected gradient vanishes only when both actions yield the same reward, i.e. when there is nothing to improve.

(e) Variance comparison: baselines $b = 3$ vs. $b = 0$

[7 marks]

Solution. The stochastic gradient with baseline is $g(A) = (R(A) - b) \partial_\theta \log \pi_\theta(A)$. Let $p = \sigma(\theta)$.

$$g(a_1) = (1-b)(1-p), \quad g(a_2) = (5-b)(-p).$$

The expected gradient (derived in part (d)) is

$$\mathbb{E}[g] = p(1-p)(1-5) = -4p(1-p).$$

This is independent of b (baseline doesn't change the expected gradient).

The second moment is

$$\begin{aligned}\mathbb{E}[g^2] &= p(1-b)^2(1-p)^2 + (1-p)(5-b)^2p^2 \\ &= p(1-p)[(1-b)^2(1-p) + (5-b)^2p].\end{aligned}$$

At $\theta = 0$: $p = \frac{1}{2}$.

$$\begin{aligned}\mathbb{E}[g^2] &= \frac{1}{4} \cdot \frac{1}{2} [(1-b)^2 + (5-b)^2] = \frac{1}{8} [(1-b)^2 + (5-b)^2], \\ (\mathbb{E}[g])^2 &= 16 \cdot \frac{1}{16} = 1, \\ \text{Var}[g] &= \frac{1}{8} [(1-b)^2 + (5-b)^2] - 1.\end{aligned}$$

$b = 0$: $\text{Var} = \frac{1}{8}(1 + 25) - 1 = 3.25 - 1 = 2.25$.

$b = 3$: $\text{Var} = \frac{1}{8}(4 + 4) - 1 = 1 - 1 = 0$.

$$\boxed{\text{Var}_{b=0}|_{\theta=0} = 2.25, \quad \text{Var}_{b=3}|_{\theta=0} = 0.}$$

The variance is **exactly zero** with $b = 3$ at $\theta = 0$. This happens because $r(a_1) - 3 = -2$ and $r(a_2) - 3 = +2$, while the score functions are $+0.5$ and -0.5 respectively. The products $(R - b) \cdot \text{score}$ are both equal to -1 regardless of which action is selected, so the stochastic gradient has no randomness.

In general, the optimal baseline (minimising variance) is $b^* = \frac{\mathbb{E}[R(A)(\partial_\theta \log \pi)^2]}{\mathbb{E}[(\partial_\theta \log \pi)^2]}$. At $\theta = 0$ where $(\text{score})^2 = p^2$ or $(1-p)^2$ are equal ($= \frac{1}{4}$), this simplifies to $b^* = \frac{p \cdot r(a_1) + (1-p) \cdot r(a_2)}{1} = \frac{1+5}{2} = 3$.

Question 3 — Control

Grid layout and shortest paths. Using BFS from each exit:

Cell	Dist. to G_L	Dist. to G_R
$(0, 0) = G_L$	0	8
$(0, 1)$	1	7
$(1, 0)$	1	7
$(2, 0)$	2	6
$\mathbf{X} = (2, 1)$	3	5
$(2, 2)$	4	4
$(1, 2)$	5	3
$(2, 3)$	5	3
$(1, 3)$	6	2
$(2, 4)$	6	2
$(0, 3)$	7	1
$(1, 4)$	7	1

(a) Conditions for the optimal policy to go right

[3 marks]

Solution. With $\gamma = 1$, step cost -1 , exit rewards $+5$ (left) and $+r$ (right):

- Optimal left path (3 steps): $-1 - 1 + (5 - 1) = 2$.
- Optimal right path (4 steps): $-1 - 1 - 1 + (r - 1) = r - 4$.

Going right is optimal iff $r - 4 > 2$, i.e.

$$r > 6.$$

(At $r = 6$ the returns tie and either exit is optimal.)

(b) Greedy policy π' after SARSA under uniform random ($r = 50$)

[5 marks]

Solution. SARSA with a uniformly random policy converges to $q_\pi(s, a)$ for the random policy.

Under the random policy with step cost -1 , the return from any state is

$$v_\pi(s) = -\mathbb{E}[T(s)] + 5 P_L(s) + 50 P_R(s),$$

where $T(s)$ is the (random) hitting time to any exit, and P_L, P_R are the exit probabilities.

The expected hitting time for a random walk on this grid is very large (random walks on finite graphs take $O(n^2)$ steps on average to hit a boundary, where n is the diameter). With only modest terminal rewards ($+5$ and $+50$), the accumulated step costs dominate, making all values very negative.

The greedy improvement selects, at each state, the action leading to the neighbour with the highest value. Since step costs dominate, values decrease with distance from the nearest exit. From the distance table: at $\mathbf{X}$, G_L is 3 steps away while G_R is 5 steps. Going left leads to $(2, 0)$ (2 steps from G_L); going right leads to $(2, 2)$ (4 steps from either exit).

$$\pi'(\mathbf{X}) = \text{left, with probability 1.}$$

(Up and down both stay at $\mathbf{X}$ due to wall/boundary, giving $q = -1 + v(\mathbf{X})$ which is worse than $-1 + v(2, 0)$ since $v(2, 0) > v(\mathbf{X})$.)

(c) Greedy policy π'' after SARSA under π'

[4 marks]

Solution. Under π' , the agent deterministically follows $X \rightarrow (2, 0) \rightarrow (1, 0) \rightarrow G_L$, collecting return $-1 - 1 + 4 = 2$.

Key observation: SARSA is *on-policy*. Under π' , the agent always goes left from X, so *only the action* $q(X, \text{left})$ gets updated. The action values $q(X, \text{right})$, $q(X, \text{up})$, $q(X, \text{down})$ remain at their initialisation of 0 (they are never experienced).

After convergence:

$$\begin{aligned} q(X, \text{left}) &\rightarrow 2, \\ q(X, \text{right}) &= q(X, \text{up}) = q(X, \text{down}) = 0. \end{aligned}$$

The greedy policy is $\pi''(X) = \text{left}$ (since $2 > 0$).

$\pi''(\mathbf{X}) = \text{left}, \quad \text{with probability } 1.$

Remark. The on-policy nature of SARSA means the agent never discovers the high-reward right exit. The policy is stuck at a suboptimal local solution: the *optimal* return from X heading right is $-3 + 49 = 46 \gg 2$.

(d) Q-learning with uniform behaviour policy

[5 marks]

Solution. Q-learning is *off-policy*: its update uses

$$q(S_t, A_t) \leftarrow q(S_t, A_t) + \alpha [R_{t+1} + \gamma \max_a q(S_{t+1}, a) - q(S_t, A_t)].$$

Under a uniformly random *behaviour* policy, all state-action pairs are visited infinitely often (the grid is communicating), so Q-learning converges to q^* .

The optimal policy satisfies: optimal return going left = 2, optimal return going right = 46. Hence $\pi'_Q(X) = \text{right}$.

In the **second round**, Q-learning is again run with the same uniform behaviour policy. Since Q-learning converges to q^* regardless of the behaviour policy (provided all pairs are visited), the result is the same.

$\pi'_Q(\mathbf{X}) = \pi''_Q(\mathbf{X}) = \text{right}.$

Contrast with SARSA: Q-learning's off-policy learning immediately discovers that the right exit is far superior ($q^*(X, \text{right}) = 46 \gg q^*(X, \text{left}) = 2$), whereas on-policy SARSA under the random policy only evaluates the random policy's own (very poor) behaviour, and then gets stuck at the left exit.

(e) Potential-based reward shaping

[7 marks]

Solution. The shaped reward is $R'_t = R_t + F(S_t, S_{t+1})$ where $F(s, s') = \gamma \Phi(s') - \Phi(s)$ with $\Phi(s) = -d(s, G_R)$ and $\gamma = 0.95$.

This is exactly the **potential-based reward shaping** (PBRs) form of Ng et al. (1999).

Theorem (PBRs). If the shaping function has the form $F(s, s') = \gamma \Phi(s') - \Phi(s)$, then the optimal policy under the shaped reward with discount γ is *identical* to the optimal policy under the original reward with the same γ .

Applying the theorem: the shaped Q-learning values converge to the optimal action-value function of the *shaped* MDP with $\gamma = 0.95$. By PBRS, the greedy policy equals the optimal policy of the *original* (unshaped) MDP with $\gamma = 0.95$.

Does $\gamma = 0.95$ (vs. $\gamma = 1$) change the optimal policy? Under the original rewards with $\gamma = 0.95$:

$$\text{Left (3 steps): } G_L = -1 + 0.95(-1) + 0.95^2(+4) = -1 - 0.95 + 3.61 = 1.66,$$

$$\text{Right (4 steps): } G_R = -1 + 0.95(-1) + 0.95^2(-1) + 0.95^3(+49) \approx 39.16.$$

Going right remains optimal. Therefore:

Yes — the greedy policy under the shaped reward equals the optimal policy

under the original reward. PBRS preserves optimality for the same γ .
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Caveat: PBRS guarantees preservation for a *fixed* γ , not across different discount factors. Here the original problem was undiscounted ($\gamma = 1$) and the shaped problem uses $\gamma = 0.95$. It is a coincidence of this MDP (the right exit is so much more rewarding) that the optimal policy is preserved. In general, changing γ can change the optimal policy.

Question 4 — Operators and Policy Improvement

The operator is

$$(\mathcal{T}q)(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a) \frac{\max_b q(s', b) + \min_b q(s', b)}{2}.$$

(a) γ -contraction?

[5 marks]

Solution. Yes, $\mathcal{T}$ is a γ -contraction w.r.t. $\|\cdot\|_\infty$.

Define $f(q, s') = \frac{1}{2}[\max_b q(s', b) + \min_b q(s', b)]$. For two value functions q, q' :

$$|f(q, s') - f(q', s')| = \frac{1}{2} \left| [\max_b q(s', b) - \max_b q'(s', b)] + [\min_b q(s', b) - \min_b q'(s', b)] \right|.$$

Using the standard inequality $|\max_a g(a) - \max_a h(a)| \leq \max_a |g(a) - h(a)|$ (and the analogous result for min):

$$\begin{aligned} |\max_b q(s', b) - \max_b q'(s', b)| &\leq \|q - q'\|_\infty, \\ |\min_b q(s', b) - \min_b q'(s', b)| &\leq \|q - q'\|_\infty. \end{aligned}$$

Therefore $|f(q, s') - f(q', s')| \leq \|q - q'\|_\infty$. Then:

$$\begin{aligned} |(\mathcal{T}q)(s, a) - (\mathcal{T}q')(s, a)| &= \gamma \left| \sum_{s'} p(s'|s, a) [f(q, s') - f(q', s')] \right| \\ &\leq \gamma \sum_{s'} p(s'|s, a) \|q - q'\|_\infty = \gamma \|q - q'\|_\infty. \end{aligned}$$

Taking the supremum over (s, a) : $\|\mathcal{T}q - \mathcal{T}q'\|_\infty \leq \gamma \|q - q'\|_\infty$. □

(b) Unique fixed point? Equal to q^* ?

[6 marks]

Solution. Since $\mathcal{T}$ is a γ -contraction on the complete metric space $(\mathbb{R}^{|\mathcal{S}||\mathcal{A}|}, \|\cdot\|_\infty)$ with $\gamma < 1$, the Banach fixed point theorem guarantees a **unique fixed point** $\hat{q}_\mathcal{T}$.

Is $\hat{q}_\mathcal{T} = q^*$? **No, in general.**

Counterexample. Consider a single-state MDP ($\mathcal{S} = \{s\}$) with two actions $\mathcal{A} = \{a, b\}$ and $\gamma = 0.9$:

$$r(s, a) = 1, \quad r(s, b) = 0, \quad p(s|s, a) = p(s|s, b) = 1.$$

Standard Bellman optimality: $q^*(s, a) = 1 + 0.9 \max(q^*(s, a), q^*(s, b))$, $q^*(s, b) = 0 + 0.9 \max(q^*(s, a), q^*(s, b))$. Assuming $q^*(s, a) > q^*(s, b)$: $q^*(s, a) = 1 + 0.9 q^*(s, a) \Rightarrow q^*(s, a) = 10$, $q^*(s, b) = 9$.

$$\text{Operator } \mathcal{T}: \hat{q}(s, a) = 1 + 0.9 \cdot \frac{\hat{q}(s, a) + \hat{q}(s, b)}{2}, \quad \hat{q}(s, b) = 0 + 0.9 \cdot \frac{\hat{q}(s, a) + \hat{q}(s, b)}{2}.$$

From the second equation: $\hat{q}(s, b) = 0.45[\hat{q}(s, a) + \hat{q}(s, b)]$, so $0.55 \hat{q}(s, b) = 0.45 \hat{q}(s, a)$, giving $\hat{q}(s, b) = \frac{9}{11} \hat{q}(s, a)$.

Substituting into the first:

$$\hat{q}(s, a) = 1 + 0.45 \hat{q}(s, a) \cdot \frac{20}{11} = 1 + \frac{9}{11} \hat{q}(s, a) \Rightarrow \hat{q}(s, a) = \frac{11}{2} = 5.5.$$

Then $\hat{q}(s, b) = \frac{9}{11} \cdot 5.5 = 4.5$.

$$\hat{q}_{\mathcal{T}} = (5.5, 4.5) \neq (10, 9) = q^*.$$

Intuition: The operator $\mathcal{T}$ averages max and min next-state values, which “drags down” the bootstrap target relative to the standard Bellman optimality operator (which uses only the max).

(c) Deterministic MDP

[4 marks]

Solution. No, $\hat{q}_{\mathcal{T}} \neq q^*$ even in deterministic MDPs (as long as $|\mathcal{A}| \geq 2$ and the actions have different values).

The counterexample from part (b) *is* a deterministic MDP: each (s, a) transitions to a unique next state (both go to s). The max and min over actions in the next state still differ ($5.5 \neq 4.5$), so the averaging changes the fixed point.

The only case where $\hat{q}_{\mathcal{T}} = q^*$ would hold is when $\max_b q(s', b) = \min_b q(s', b)$ for all reachable s' (i.e. all actions in every successor state have equal value), or trivially when $|\mathcal{A}| = 1$.

(d) Does $\pi_{\max}(a|s) = \arg \max_a [q_{\pi_1}(s, a) + q_{\pi_2}(s, a)]$ improve on either?

[5 marks]

Solution. Answer: (ii) — $\pi_{\max}$ need not improve on either π_1 or π_2 .

Counterexample. Consider three states s_1, s_2, s_3 with three actions a, b, c and $\gamma = 0.9$:

State	Action	Reward	Next state
s_1	a	10	s_1
s_1	b	0	s_1
s_1	c	0	s_3
s_2	a	0	s_2
s_2	b	10	s_2
s_2	c	0	s_3
s_3	any	7	s_3

π_1 : always takes a . $v_{\pi_1} = (100, 0, 70)$.

π_2 : always takes b . $v_{\pi_2} = (0, 100, 70)$.

These are incomparable: $v_{\pi_1}(s_1) > v_{\pi_2}(s_1)$ but $v_{\pi_2}(s_2) > v_{\pi_1}(s_2)$.

Computing $q_{\pi_1} + q_{\pi_2}$ at s_1 :

$$\begin{aligned} q_{\pi_1}(s_1, a) + q_{\pi_2}(s_1, a) &= 100 + 10 = 110, \\ q_{\pi_1}(s_1, b) + q_{\pi_2}(s_1, b) &= 90 + 0 = 90, \\ q_{\pi_1}(s_1, c) + q_{\pi_2}(s_1, c) &= 63 + 63 = 126 \leftarrow \text{highest!} \end{aligned}$$

By symmetry, action c also wins at s_2 . So $\pi_{\max}$ takes c at both s_1 and s_2 , sending the agent to s_3 where it earns $7/(1 - 0.9) = 70$.

$$v_{\pi_{\max}}(s_1) = 0 + 0.9 \times 70 = 63 < 100 = v_{\pi_1}(s_1). \quad 55$$

$$v_{\pi_{\max}}(s_2) = 63 < 100 = v_{\pi_2}(s_2). \quad 55$$

$\pi_{\max}$ fails to improve on either policy. Maximising the *sum* of q -functions can select a “compromise” action (c) that has high combined value due to s_3 ’s moderate utility, but is individually suboptimal for each policy.

(e) A policy that provably improves on both

[6 marks]

Solution. Construction. Define

$$\pi'(s) = \arg \max_a \max(q_{\pi_1}(s, a), q_{\pi_2}(s, a)).$$

That is, at each state pick the action whose *better*-policy q -value is highest (the pointwise max of the two q -functions, then greedy over actions).

Proof that $v_{\pi'}(s) \geq \max(v_{\pi_1}(s), v_{\pi_2}(s))$ **for all** s .

Define $v'(s) = \max(v_{\pi_1}(s), v_{\pi_2}(s))$.

Step 1. Show $(T^{\pi'} v')(s) \geq v'(s)$ for all s .

Let $a^* = \pi'(s)$. Since the MDP model gives us the identity $r(s, a) + \gamma \sum_{s'} p(s'|s, a) v'(s') \geq q_{\pi_j}(s, a)$ for each j (because $v'(s') \geq v_{\pi_j}(s')$):

$$\begin{aligned} (T^{\pi'} v')(s) &= r(s, a^*) + \gamma \sum_{s'} p(s'|s, a^*) v'(s') \\ &\geq \max_j q_{\pi_j}(s, a^*) && \text{(since } v' \geq v_{\pi_j}) \\ &= \max_a \max_j q_{\pi_j}(s, a) && \text{(by def. of } a^*) \\ &\geq \max_a q_{\pi_i}(s, a) \geq v_{\pi_i}(s), \quad \forall i. \end{aligned}$$

Taking the max over i : $(T^{\pi'} v')(s) \geq \max_i v_{\pi_i}(s) = v'(s)$.

Step 2. By monotonicity of $T^{\pi'}$, applying inductively:

$$v' \leq T^{\pi'} v' \leq (T^{\pi'})^2 v' \leq \dots \leq (T^{\pi'})^n v' \xrightarrow{n \rightarrow \infty} v_{\pi'}.$$

Hence $v_{\pi'}(s) \geq v'(s) = \max(v_{\pi_1}(s), v_{\pi_2}(s))$ for all s . □

Verification on the counterexample from (d): At s_1 : $\max(q_{\pi_1}(s_1, a), q_{\pi_2}(s_1, a)) = 100$; $\max(q_{\pi_1}(s_1, c), q_{\pi_2}(s_1, c)) = 63$. So π' picks a at s_1 . By symmetry π' picks b at s_2 . Then $v_{\pi'}(s_1) = 100$ and $v_{\pi'}(s_2) = 100$, improving on both.

END OF SOLUTIONS